Sahaj Bhadja

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Summary

Full-stack software engineer pursuing an MSCS at USC with 2+ years building **TypeScript/React interfaces**, **Python backend systems**, and cloud-backed data workflows, with hands-on experience in scalable pipelines, CI/CD, and production issue resolution.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL patterns, and improving responsiveness for analytics and log-style workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving reliability, release confidence, and observability across recurring production data-processing pipelines.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **responsive full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable backend services and polished interfaces while improving delivery predictability and implementation ownership.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations through concurrent execution, CI/CD-backed releases, and cost-aware debugging.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation, faster diagnostics, and high-confidence benchmarking while improving signal quality for model-driven workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for user-facing applications.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: TypeScript, JavaScript, Python, Java, C++, SQL, Bash
Frontend: React, Next.js, responsive UI development, component-driven interfaces
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, scalable services, ETL pipelines
Cloud & Infrastructure: AWS, Docker, Kubernetes, Jenkins, Git, CI/CD, cloud-backed workflows
Data: PostgreSQL, MySQL, MongoDB, SQL Server, data processing, log-style analytics workflows
Engineering: unit testing, integration testing, technical documentation, production support, code reviews, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a **document ingestion and retrieval platform** over 50K+ records, enabling semantic search and summarization with low-latency indexing pipelines and scalable metadata-backed storage workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid*

- Built a **responsive TypeScript platform** supporting secure bank linking, real-time transaction visibility, and advanced filtering across 1M+ transactions for 50,000+ users through scalable data-backed workflows.